AOD-Net: A Spatio-Temporal Segmentation Approach for Abandoned Object Detection in Videos

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Abstract. Monitoring of forgotten and abandoned objects automatically is a crucial part of public safety. However, conventional methods of identification face problems while detecting abandoned objects on videos. In this work, we put forward a novel automated spatio-temporal framework for abandoned object identification. Due to lack of available annotated data, we have curated our own dataset consisting of 26 videos for abandoned object detection. We combine the rolling median background estimate and the current video frame by using the 6-channel early fusion method. Then the spatial identification network uses a U-Net architecture that has ResNet-18 as its encoder. The identity tracking Hungarian algorithm is used to link those frames together where regions of abandoned objects have already been identified, this provides temporal consistency and eliminates false positives which were caused due to changes in light or shadow, which is then followed by a dense optical flow technique to verify motion. We use a deterministic state machine which relies on time to develop an effective mechanism for abandoned object detection that minimizes false positives. We exhaustively compare our model against competing baseline methods resulting in a precision of 0.746, an overall F1 score of 0.744, and a detection accuracy of 90.2%. We also validated by testing our model on the publicly available ABODA dataset.

1 Introduction

Automated video surveillance now factors into public safety, traffic regulation, and smart city infrastructure. One of the harder problems in this space is Abandoned Object Detection (AOD): telling apart objects that stay static for a long time, like unattended luggage or suspicious packages at airports, from temporary cases such as standing passengers or belongings picked up again quickly. Computer vision and deep learning have advanced, but dynamic environments still make it hard to detect and track an object's full lifecycle. Traditional AOD methods extensively rely upon methods such as background subtraction algorithms, frame differencing, or simple tracking-by-detection paradigms. These techniques often struggle with changes in lighting conditions, occlusions, and crowded scenes. Furthermore, a major limitation in the functioning of these systems is the “eternal alarm" problem. Even if these detecting systems correctly

detect an abandoned object, their standard frameworks will keep the alarm ringing forever. This continuous flagging blocks the system from merging the object with the background. Thus making the surveillance feed and the detection system useless for detecting any other anomalies within the region.

To address these aforementioned challenges, this paper introduces the method of *AOD-Net*: a novel spatio-temporal segmentation pipeline catering to accurate and automatic Abandoned Object Detection. To measure our framework's performance, we curate a new custom dataset that consists of 26 surveillance videos. We propose a multi-stage architecture that reduces the gap between spatial segmentation and temporal object tracking. Our spatial detection framework not only performs simple single-frame analysis but also incorporates a background modeling technique which estimates the normal state of the scene by using a rolling median of recent frames on the surveillance video. The current video frame is then stacked with the calculated background estimate creating a 6-channel input tensor. This tensor is then fed to a U-net based architecture which has ResNet-18 as its backbone encoder. This extracts the hierarchical visual features to output a per-pixel probability mask which isolates static foreign objects.

However, isolating a temporary and permanent abandoned object cannot be achieved by spatial detection alone. Hence, the Hungarian algorithm is then used to link detected regions across frames thus ensuring temporal consistency. This is done to assign and maintain a persistent tracking ID for each object over time. Then we run a dense optical flow check to confirm if the tracked objects/regions are genuinely stationary or not. This effectively removes false positives in the background model caused by dynamic shadows and lighting changes. Finally, a deterministic temporal gating mechanism is used to track the lifecycle of these detected regions.This mechanism works on the principle of a timing rule based on Persistence time ($T_{persistent}$) and Cooldown time ($T_{cooldown}$), consistent as used in persistent scene change detection[1]. Intuitively $T_{persistent}$ is the time an object has to be at rest before it is marked as abandoned, while $T_{cooldown}$ is the time after which a flagged object is absorbed into the background. The gating pathway reports an object as abandoned if it is found to be in the same state for a duration greater than $T_{persistent}$. This object will be marked in white color in the generated output mask. Alternatively we implement a background absorption mechanism to solve the "eternal alarm" problem. If the detected object is found to be stationary for an extended period of $T_{cooldown}$ time, it is deleted from the list of potentially abandoned objects, reverting its output mask back to black. We refer to this transition as "absorption": the object is not just un-flagged but the system's model considers it as normal background, so the object's continued presence is not considered an anomaly any longer. In short, the mask of an object is black while the system is still counting the time $T_{persistent}$. It becomes white once the system marks it as abandoned, and returns to black when it gets

[1] PSCDL 2026 Challenge: `https://www.vehant.com/the-persistent-scene-change-detection-and-localization-pscdl-2026/` and `https://ncvpripg2026.lnmiit.ac.in/challenges`.

absorbed at $T_{cooldown}$. This allows the machine to redirect its attention to new events that may potentially trigger an anomaly.

To summarise, our key contributions in this work are as follows:

1. We introduce a 26-video, pixel-level annotated dataset for abandoned object detection. Unlike existing benchmarks such as ABODA, which provide only bounding-box annotations, ours span seven object categories, a 65× range in object footprint, and multi-object scenarios with temporal overlap, paired with a spatial detection framework that uses 6-channel background fusion and a U-Net with a ResNet-18 encoder.
2. We combine the Hungarian algorithm for identity matching with dense optical flow for motion verification. Jointly it identifies false positives from shadows and lighting changes.
3. Leveraging persistence/cooldown timing framework, we implement a deterministic state machine using $T_{persistent}$ and $T_{cooldown}$ thresholds to achieve dynamic background absorption, solving the eternal alarm problem that causes conventional AOD systems to flag a resolved event indefinitely.
4. We benchmark AOD-Net against five baseline competing methods – CNN-based, transformer-based, state-space-model-based, and self-supervised representation learning – and validate its generalization through a zero-shot evaluation on the public ABODA benchmark, with no fine-tuning.

2 Related Work

2.1 Traditional Abandoned Object Detection and Background Modeling

Traditionally Abandoned Object Detection (AOD) techniques were primarily based on frame differencing and background subtraction. Prior studies such as the survey by SanMiguel and Cavallaro et al. [9] points out that although traditional methods are computationally efficient, they struggle with sudden and rapid changes in illumination, occlusions, and dynamic backgrounds. Xiong et al. [14] recently achieved a breakthrough in system robustness by introducing a long-term adaptive background modeling technique. This technique continuously updates background estimate in response to the environmental changes. To filter out the transient noise, Zhou and Xu [18] recently introduced a dual-background modeling technique combined with deep learning. Inspired by such advancements, we introduced a background estimation method that utilizes rolling median. A key feature of our approach is that it integrates background subtraction with rolling median technique, which are then fed into our 6-channel early fusion system.

2.2 Deep Learning and Semantic Segmentation in Surveillance

The shift from traditional background subtraction to Convolutional Neural Networks (CNNs) and other deep learning techniques has greatly enhanced spatial object detection. Architectures such as the Mamba U-Net framework [13] for convolution-free segmentation or a ResNet framework proposed by He et al. [4] are considered to be vital in the methods to determine state of the art pixel-wise

predictions. Real-time bounding box detectors, such as YOLO26n-seg [5], have already become benchmarks in current video surveillance systems.

Abandoned object detection (AOD) techniques have been actively researched with many discoveries recently. Xu et al. [15] introduced a ROAD standard proving the need for intelligent and automated city infrastructure. Additionally Yilmazer and Karakose [16] introduced a framework that combined tracking features and spatial detectors with dynamic prompting.

The field of AOD is continually growing with latest and sophisticated technologies. Recent developments consists of a two stage spatiotemporal CNN-YOLOv9 model [11] and modified Mask R-CNN techniques for real-time detection at public places [12]. These new developments also recognize the importance of training-less techniques [6]. AOD has progressed so much that it now also includes technologies other than computer vision, such as the use of distributed ultra-weak fiber Bragg grating (UWFBG) sensors for real-time road monitoring [7]. We introduce a spatial network based on a U-Net and ResNet-18 as its backbone encoder, which results in a per-pixel probability map rather than utilizing bounding box baselines.

2.3 Multi-Object Tracking and Temporal State Management

Spatial detection for AOD is essential, but it alone does not suffice. The system must also be able to distinguish between an abandoned object and simply a pedestrian stopping for a brief time. Hence, temporal tracking is also essential. The Hungarian algorithm is considered the theoretical basis of bipartite data association. This increased the popularity of modern tracking-by-detection systems, such as ByteTrack [17]. To verify motion and eliminate false positive results due to illumination and shadow changes, many techniques based on dense optical flow have been developed, such as polynomial expansion by Farnebäck [3].

To solve the "eternal alarm" issue, temporal consistency rules are essential. The Abandoned Object Dataset (ABODA) applies this principle by utilizing temporal consistency and tracking for AOD [8]. We take this a step further by incorporating a deterministic temporal state machine. Using Hungarian matching and optical flow verification, we introduce a background-updating mechanism governed by the $T_{persistent}$ and $T_{cooldown}$ thresholds introduced in Section 1, which folds abandoned or stationary objects back into the background once flagged.

3 Proposed Methodology

3.1 Background Modeling and Feature Fusion

In order to successfully segregate items that are not supposed to be in the scene, the system creates a dynamic reference of the surroundings as illustrated in Figure 1. The system keeps a rolling buffer of the latest frames, updated for every new incoming frame. The buffer has a track of about three seconds. The estimated background at time t is computed as the median of the pixel by pixel and channel by channel values over the frames held in this rolling buffer:

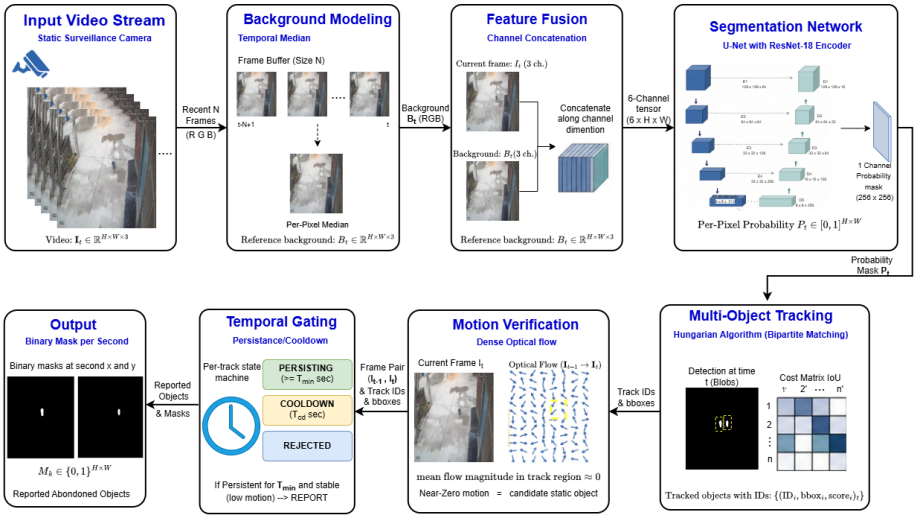


Fig. 1. Overall Architecture of the Proposed AOD-Net Pipeline .

$$B(x, y, c, t) = \text{median}\{I(x, y, c, t-k) : k = 0 \dots N-1\} \tag{1}$$

Here, I symbolizes the input frame, (x, y) denotes pixel coordinates in the spatial domain, c is for RGB color channels and N is the length of the buffering window. The usage of median in place of mean allows for effective elimination of transient motion from the background, e.g. the pedestrians or passing vehicles allowing the true static background to be recovered. Once the background estimate B_t is calculated, I_t and B_t are stacked along the channel axis which creates a six-channel input tensor X_t.

$$X_t(x, y) = \Big[I_t(x, y), B_t(x, y)\Big] \in \mathbb{R}^{H\times W\times 6} \tag{2}$$

Here, we have $I_t(x, y)$ and $B_t(x, y)$ which represent the full 3 channel RGB pixel vectors of the current frame and background estimate respectively, as per $I(x, y, c, t)$ in Equation 1 in which we are leaving the c out for simplicity; the two are put together along the channel axis to form the 6 channel tensor X_t.

The six-channel tensor provides the network with a pixel-aligned cue indicating what changed relative to the estimated background. Without this cue, the network would need to learn such changes from single frame appearance alone, which would increase the burden of temporal reasoning on the network.

3.2 Spatial Segmentation Network: U-Net with ResNet-18

The segmentation network presented here, illustrated in Figure 2, uses a pretrained ResNet-18 [4] encoder and inputted with the six-channel input tensor, with a design that includes five stages of downsampling. At each stage, the encoder reduces the spatial resolution while increasing the number of feature channels. This progressive downsampling is what allows the encoder to develop

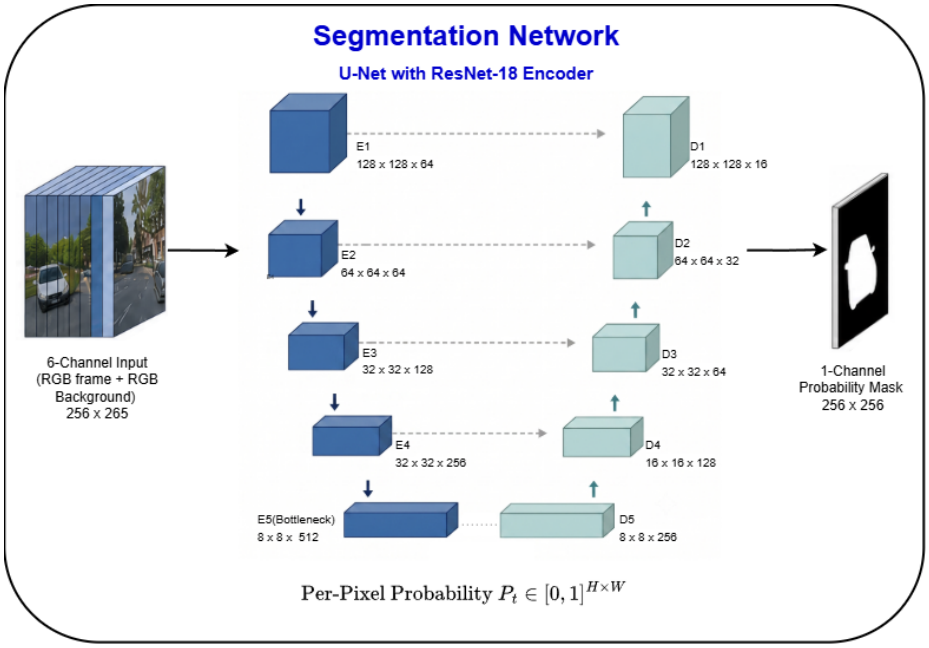


Fig. 2. Detailed architecture of the Spatial Segmentation Network.

increasingly abstract, high-level representations of the scene, so that even at a spatial resolution of just 1/32 that of the original input, the output has grown to 512 feature channels.

Pixel-level masks require precise spatial information, so the architecture uses a mirrored decoder. Over five stages, it restores the bottleneck features to their original resolution. The channel count is halved at each step—256, 128, 64, 32, and finally 16. At every stage, a skip connection combines the upsampled output with a high-resolution feature map from the matching encoder stage, helping to retain fine details. Finally, a 1×1 convolution at the final stage reduces the output to a single channel and a per pixel probability map is produced which in turn indicates which pixels are part of a static or foreign object.

The network is then trained with a combined loss function, $\mathcal{L}_{total}$, defined as the sum of Binary Cross-Entropy (BCE) loss and Dice loss. BCE stabilizes pixel-wise classification, while Dice loss encourages geometric overlap with the ground-truth mask Y:

$$\mathcal{L}_{total} = \mathcal{L}_{BCE}(\hat{Y}, Y) + \mathcal{L}_{Dice}(\hat{Y}, Y) \tag{3}$$

Here, the predicted probability mask is denoted as $\hat{Y}$. the individual loss components are defined as:

$$\mathcal{L}_{BCE} = -\frac{1}{H \times W} \sum_{i=1}^{H \times W} [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \tag{4}$$

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum_{i=1}^{H \times W} y_i \hat{y}_i}{\sum_{i=1}^{H \times W} y_i + \sum_{i=1}^{H \times W} \hat{y}_i + \epsilon} \tag{5}$$

ϵ is used as a small smoothing constant to prevent division by zero. Here, y_i and $\hat{y}_i$ denote the ground-truth and predicted values in pixel i, respectively.

3.3 Multi-Object Tracking and Motion Verification

The spatial network generates an initial probability map that requires thresholding in order to distinguish connected components, which are known as blobs. The important thing to remember is that once the blobs have been formed, they need to be tracked through the next video frames by preserving their identity so that we can trace the persistence of the object over time. Hence, we apply the Hungarian algorithm for bipartite graph matching, whereby we link the previously identified blobs with the blobs that have been detected at the instances or objects present. The cost matrix in the Hungarian algorithm is calculated as the distance between the centers of each pair of blobs. A candidate match farther than 45 pixels apart is rejected; when a track has no valid match within this range, it is instead registered as a new, separate object.

$$C_{i,j} = \|\text{centroid}(d_i) - \text{centroid}(t_j)\|_2 \tag{6}$$

Here, d_i is the i-th blob found in the current frame, while t_j is the j-th registered track. Thus $C_{i,j}$ represents the cost of linking blob d_i to track t_j. Assignments do not exceed a maximum distance limit, so pairs lying beyond this threshold are rejected preventing incorrect merging.

Nevertheless, temporal duration alone is not enough for confirming an abandoned object: slowly shifting shadows or lighting changes may cause a track to reach a valid duration without corresponding to any real static object. To solve this, we use dense optical flow based on the Farnebäck method [3], computing the mean flow magnitude within each tracked blob's mask per frame. The mean flow magnitude is calculated for each tracked object. A track is only considered motion verified when 80% of its past values is classified as motion-static.

$$\mu_{flow}(A_k) = \frac{1}{|A_k|} \sum_{(x,y) \in A_k} \sqrt{u(x,y)^2 + v(x,y)^2} \tag{7}$$

Here, A_k denotes the set of pixels of tracked blob k's mask, $u(x, y)$ and $v(x, y)$ are the horizontal and vertical components of the optical flow vector at pixel (x, y). A track is considered motion verified if enough of its past values shows μ_{flow} near zero, indicating negligible motion.

3.4 Temporal Logic and State Machine

All recorded tracks are passed through a deterministic state machine, governing how long it must persist before being marked as abandoned and how long it persists afterward before it is absorbed in the background. This is done using the two parameters – Persistence Threshold ($T_{persistent}$) and Cooldown Period ($T_{cooldown}$).

t_0 denotes the time at which a track first becomes motion-verified, rather than the time at which it is first detected. The track state $S(t)$ is then classified as:

$$S(t) = \begin{cases} \text{TRACKING}, & t_0 \leq t < t_0 + T_{persistent} \\ \text{ALERT (Abandoned)}, & t_0 + T_{persistent} \leq t < t_0 + T_{persistent} + T_{cooldown} \\ \text{ABSORBED}, & t \geq t_0 + T_{persistent} + T_{cooldown} \end{cases} \tag{8}$$

For an object to be considered “abandoned" it must have surpassed the $T_{persistent}$ threshold and been confirmed stationary through dense optical flow. In order to resolve the concern of “eternal alarm" that comes with traditional AOD systems, the state machine uses the $T_{cooldown}$ rule. If an object shows no movement for an additional $T_{cooldown}$ seconds after its first detection, the system recognizes the object as part of the environment permanently. The tracking of the object stops and the alert message is suppressed, thus the object is absorbed in the background model.

4 Experimental Results

4.1 Experimental Setup and Implementation Details

Each video in our dataset consists of two timing parameters $T_{persistent}$ and $T_{cooldown}$. These are unique to each video instead of a fixed constant threshold. $T_{persistent}$ threshold ranges from 10 to 40 seconds with a mean of 28 seconds and $T_{cooldown}$ ranges from 15 to 60 seconds with a mean of 43 seconds. A lower learning rate is used for the pre-trained ResNet-18 encoder (5×10^{-6}) which fine-tunes its low level features slowly. At the same time a higher learning rate (5×10^{-5}) is used for the randomly initialized decoder which causes it to learn its features faster. We use the combined loss as defined in Eq. 3 which is a combination of binary cross-entropy and Dice loss. We reduce learning rate when validation F1 score plateaus and also we utilize early stopping criteria while training after eight epochs of no improvement. With this schedule, the final model converges quickly. The best validation epoch is epoch 4 out of 12, reached well before early stopping is activated. Given the relatively modest size of the training set, that result is not unexpected. Before they are combined into the six-channel input tensor, both the current frame and the rolling-median background estimate are resized to 256×256[2].

[2] Code will be released upon publication.

The segmentation network is trained with a fixed random seed controlling weight initialization and batch shuffling, with GPU convolution algorithms pinned to deterministic mode. The final model is trained in full 32-bit precision rather than mixed precision as explained in Section 4.3.1.

Training labels in our system are created from manually painted object masks while their video interval and threshold annotations are processed by a preparation script rather than manual frame by frame labeling. We determine each object's true appearance time to be the start of its first annotated interval minus $T_{persistent}$. We produce per-frame training labels, which are the union of all objects present at a given timestamp based on their individual appearance times.

4.2 Dataset Details

We introduce a custom 26-video dataset for abandoned object detection, in which each fixed-camera surveillance clip begins with a clean scene and later consists of an object introduced into the scene. In contrast to detection benchmarks that only use bounding-box annotations, these videos incorporate pixel-level ground-truth masks that indicate the position of the abandoned object for as long as it is present as well as during the subsequent cooldown periods. The video lengths vary from 49 to 354 seconds (average 143 seconds).

Object sizes are quite different, ranging from 0.21% to 13.88% of the frame, which provides about 65× in area; thus, detecting small objects seems to be a major issue for most of the videos. Table 1 shows the translation of this into the seven object categories of the dataset.

Table 1. Object category diversity and per-category frame footprint, computed from ground-truth mask pixel counts.

Category	**# Masks**	**Min % Frame**	**Max % Frame**	**Mean % Frame**
Bike	11	0.85	5.66	2.68
Bucket	6	0.21	0.76	0.48
Car	8	∼0	13.88	2.66
Chair	2	1.00	1.12	1.06
Gas cylinder	2	0.55	0.79	0.67
Scooter	3	1.12	3.85	2.42
Suitcase	2	0.68	2.37	1.53

A total of six videos from the collection of 26 (23%) featured two objects presented at different moments of time. One of the videos illustrates an instance when the first object blocks the view of the second. Altogether, the videos feature 16 different resolutions of masks ranging from 585×1035 to 1918×876. The objects occupy various positions in the video frames including the center, edge, and corners and don't always gather in the center of the frame. For final evaluation, 21 videos are used for training, and 5 are held out for validation.

4.3 Quantitative Analysis

Table 2 outlines the final performance of our model under two different evaluation criteria. The frame-by-frame evaluation computes pixel-level precision, recall,

F1, and Intersection over Union (IoU) at a fixed threshold of 0.5, on held-out validation frames evaluated during training. The full-pipeline (video-by-video) evaluation instead measures the performance of the complete system end to end, from segmentation through tracking, motion verification, and temporal gating. In this way, we are able to compare the predicted masks to the corresponding ground-truth intervals.

Table 2. AOD-Net performance under the per-frame and full-pipeline evaluation protocols, on the 26-video custom dataset.

Metric	**Per-Frame Validation**	**Full-Pipeline (All Videos)**
Precision	0.645	0.746
Recall	0.410	0.769
F1-score	0.502	0.744
IoU	0.335	0.652
Second-level detection accuracy	-	90.2%

The discrepancy between the per-frame F1 score of 0.502 and the overall F1 score of 0.744 reflects the different evaluation protocols rather than a contradiction between the scores. The per-frame score is achieved based on the pixel output of the segmentation network, estimated only from the validation frames. The overall score benefits from the use of tracking, motion verification, and temporal gating, which eliminate false alerts, but it also incurs a penalty whenever the system fails to detect the object in a video entirely.

To compare AOD-Net against other competing methods, we evaluate five baseline algorithms: YOLO26n-seg [5], which is a general-purpose pre-trained single-frame object detector; DeepLabV3 [2], which is a classical CNN-based semantic segmentation architecture; Mamba U-Net [13], which employs the Mamba state-space-model framework instead of convolutional encoding; I-JEPA [1], used here with a frozen self-supervised pre-trained encoder and trained decoder head; and a VADMamba-inspired [10] frame-prediction method adapted from video anomaly detection. All of these methods except AOD-Net were not trained on the six-channel frame-background tensor as discussed in Section 3; they work with the raw current frame, except VADMamba, which operates on frame-prediction error instead.

Table 3. AOD-Net vs. five baseline methods, fair same-split, per-frame pixel-level comparison.

Model	**Precision**	**Recall**	**F1**	**IoU**
AOD-Net (Ours)	0.645	0.410	0.502	0.335
YOLO26n-seg (pre-trained)	0.308	0.618	0.411	0.259
VADMamba-inspired	0.158	0.421	0.230	0.130
I-JEPA (frozen encoder + decoder)	0.515	0.226	0.315	0.187
Mamba U-Net	0.408	0.285	0.336	0.202
DeepLabV3	0.734	0.274	0.399	0.249

YOLO26n-seg gets higher recall than AOD-Net, but YOLO26n-seg's advantage comes at a cost in precision, as it has a much higher number of false positives and lower F1 and IoU scores despite detecting more true objects. DeepLabV3 has the highest precision in Table 3 (0.734) but one of the lowest recall scores (0.274), behind only I-JEPA (0.226), which means that a CNN segmentation model gives priority to a conservative approach in case no background information is available. The precision-recall performance of Mamba U-Net (0.408 and 0.285, respectively) indicates that a state-space-model backbone does not resolve this limitation either. The I-JEPA technique, which uses a frozen, self-supervised encoder, has the second-best precision (0.515) but low recall (0.226), which is in accordance with representations developed for general semantic tasks and not for the precise localization of small static objects. The frame-prediction approach of the VADMamba algorithm gives the weakest result (F1 score of 0.230), as a motion-anomaly signal designed to capture moving irregularities transfers poorly to a task that is fundamentally about detecting the absence of motion. Thus, the poor performance shown by the previous four baseline models, shows the small-object instability observed in AOD-Net (Section 4.3.1) is an attribute of the dataset itself and not specific to our architecture, as the difficulty persists across CNN, transformer, and state-space-model baselines alike.

Beyond the five-baseline comparison on our own dataset, we test AOD-Net's ability to generalize to unseen data by evaluating it, without any fine-tuning, on the publicly available ABODA benchmark [8]. Out of ABODA's 11 videos, one was excluded: it shows an extremely crowded public concourse where the abandoned object occupies a very small fraction of the frame relative to the surrounding pedestrian density, making it infeasible to construct a reliable ground-truth mask. As with our custom dataset, ABODA does not come with ready-made ground-truth masks, so masks for the remaining 10 videos were constructed manually. AOD-Net is then evaluated on these 10 videos using the identical full-pipeline protocol, decision threshold, and per-video $T_{persistent}/T_{cooldown}$ configuration approach used to produce Table 2.

Table 4. Results on 10 ABODA benchmark videos.

Video	IoU	F1	Detection Accuracy
sample_001	0.588	0.740	0.802
sample_002	0.835	0.910	0.957
sample_003	0.905	0.950	1.000
sample_004	0.807	0.893	0.989
sample_005	0.895	0.945	1.000
sample_006	0.956	0.977	1.000
sample_007	0.000	–	0.920
sample_008	0.952	0.976	1.000
sample_009	0.234	0.379	0.917
sample_010	0.000	–	0.669
Mean (10 videos)	**0.617**	**0.677**	**0.925**

In the custom ABODA dataset, a video with no valid detections scores 0 rather than being excluded, and an undefined F1 (0 precision and 0 recall) is likewise treated as 0 and shown as "–". Eight of the ten videos reach good pixel-level accuracy (IoU $\geq$ 0.58, while others above 0.9). The remaining two, `sample_007` and `sample_010`, demonstrate the failure of detecting small-objects as we had observed in our own data set (Section 5). However, both these videos still give real detection accuracy (0.920 and 0.669) despite zero spatial overlap, thus demonstrating that the temporal gate triggers even though segmentation fails to locate the object.

4.3.1 Ablation Study We experimented on three variants of the final model (as per Table 6) based on different hypotheses regarding the pipeline improvements on our curated dataset. The three variants did not produce better outcome than the final model.

Reduced Encoder Depth.

The network struggles to detect small objects (which occupy less than 1% of the frame). This is because of heavy downsampling (32×) in the encoder, which can shrink small object's information before it reaches the network's most compressed layer. Reducing the depth of the encoder preserves more spatial detail of individual frames. However, the analysis of the entire pipeline showed mixed results (Table 5). One video that had blank detections became detectable but two previously strong videos degraded, as decreasing the number of stages from 5 to 4 reduces the semantic context which makes the network prone to false positives from shadows or textures.

Table 5. Effect of reduced encoder depth.

Configuration	Mean Pixel F1	Mean IoU	# Videos with Blank Detections
Encoder depth 5 (final)	0.744	0.652	2
Encoder depth 4	0.471	0.378	7

Gradient Clipping. Gradient clipping was tested as a possible solution for reduced-encoder (depth 4) instability. It could have been the reason that spikes in the gradient were causing this instability. But even both thresholds (1.0 and 5.0) produced worse results as compared to the case with unclipped gradient training. In early training, the weights of the decoder are randomly assigned and need large updates to learn useful features. Clipping caps every update at a fixed size, so it does not just block the harmful spikes but also blocks the large updates needed. This slowed learning instead of stabilizing it.

Mixed-Precision (AMP) vs. Full Precision.

During gradient clipping testing, we noticed that the reduced-endoer (depth 4) produced infinite and NAN gradient values when trained in 16-bit floating point (fp16). This likely happens because fp16 can represent a limited range on numbers. When a network becomes confident in a prediction, loss calculation can grow very large. This value can exceed the range of fp16, resulting in infinite or

[2] ABODA dataset: `https://github.com/kevinlin311tw/ABODA`

NAN values. Switching to 32-bit precision fixed this problem, but the accuracy stayed poor, with the F1 score showing just ≈ 0.05. This shows that the fp16 issue and the accuracy issue were two different problems. Table 6 summarizes all three variations; none improved on the baseline.

Table 6. Performance metrics for each ablation variation.

Configuration	Precision	Recall	F1	IoU	Accuracy	Val Loss
Full precision / AMP disabled (encoder depth 4)	0.129	0.032	0.051	0.026	0.994	1.009
Gradient clipping (encoder depth 4)	0.231	0.058	0.093	0.049	0.994	0.997
Encoder depth 4	0.750	0.264	0.391	0.243	0.996	0.836
Final configuration (encoder depth 5) (AOD-Net)	0.645	0.410	0.502	0.335	0.991	0.786

4.4 Qualitative Analysis

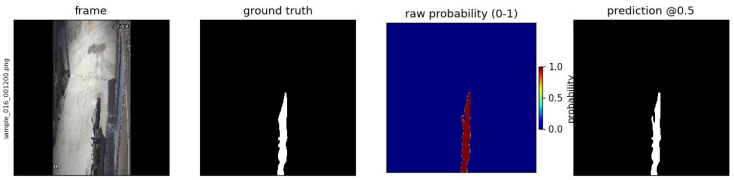


Fig. 3. An excellent detection example: the predicted mask is very close to the object's ground truth with high values of raw probability directly over the object.

Figure 3 shows the case where the predicted mask closely resembles the ground truth capturing its shape rather than a rough approximation. As Figure 4 shows, the detected area (highlighted in red) follows the vehicle reliably across different timestamps, with no flickering. Figure 5 is a strong detection example in a more complex case. A second abandoned object is introduced midway through the video, the ground-truth region grows to include it, and the model's prediction masks indicate that the model tracks this change rather than staying fixed on the first object alone. As demonstrated in Figure 6, there are cases where the system fails to produce a usable detection, consistent with the small-object blindness limitation.



Fig. 4. Another strong example: a parked vehicle tracked over 45 seconds with a full-pipeline IoU of 0.88.

5 Conclusion

This work proposed the development of AOD-Net which we put forth as a spatio-temporal framework for abandoned object detection. We reported F1 score of 0.744 and a 90.2% detection accuracy which we achieved on our custom 26 video dataset. Also, we outperformed a pre-trained YOLO26n-seg baseline, along with

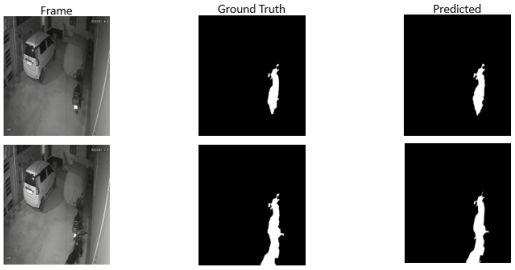


Fig. 5. A scene where two abandoned objects are introduced sequentially. First a scooter (top row) and then a motorbike (bottom row) parked behind it. At both the stages, the predicted mask closely aligns with the ground truth.

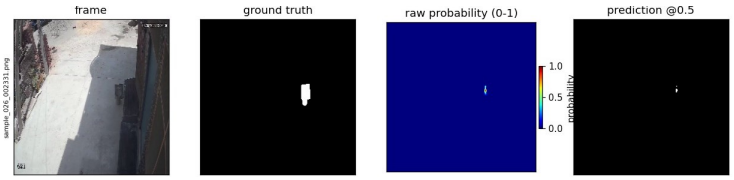


Fig. 6. A weak detection example: the object can be easily identified in the ground truth, but both the raw predicted probability and the resulting mask show almost no evidence of any abandoned object.

four other baselines – DeepLabV3, Mamba U-Net, I-JEPA, and a VADMamba approach. We noted that without any fine tuning, the pipeline does in fact generalized to the public ABODA benchmark. We achieved F1 score of 0.677. The final model reports zero detections in two videos, these videos consist of the objects that occupy less than 1% of the frame. We tested three different methods to address this issue but there was no improvement. Also we noted that for dynamic background scenes that have continuous street level activity violate the rolling median background model assumption that a pixel is unoccluded for at least half the buffer window, an issue that other background subtraction methods also generally have.

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